Quiz 10 Reflection

SCIE 001 Physics

Noah Virjee
45515863

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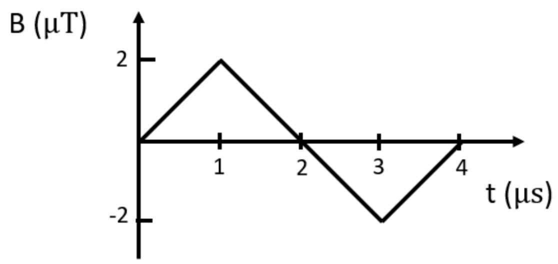
Contents

1. Magnetic induction

1.1. Specify Question

Question 5: Magnetic induction

The magnetic field through a circular loop of radius $16\,\mathrm{cm}$ varies with time as shown below. The field is perpendicular to the loop. What is **magnitude** of the induced emf in the loop at $2\,\mu s$.



$|\mathrm{emf}| =$ 2 N ✕ 0%

Correct answer

$|\mathrm{emf}| =$ `0.1608495438637974 volt`

1.2. Diagnosis Phase

I didn't know what Electromotive Force was or how to calculate it so I just took the magnitude of the slope of the line and added Newtons because it said "Force"

Words: 31

1.3. Correction Phase

Electromotive force is defined as the negative change in the magnetic flux per unit time.

$$\varepsilon = -\frac{\mathrm{d}\Phi_m}{\mathrm{d}t}$$

In this case, our flux is

$$\Phi_m = \int_s \vec{B} \cdot \hat{n}\,\mathrm{d}A$$

Since the field is uniform and perpendicular to the loop of wire, and our area is just the area of the loop:

$$\Phi_m = \pi r^2 B$$

Looking at our emf:

$$|\varepsilon| = \left|-\frac{\mathrm{d}}{\mathrm{d}t}(\pi r^2 B)\right| = \pi r^2 \left|\frac{\mathrm{d}B}{\mathrm{d}t}\right|$$

We know that at 2 μs, $\frac{\mathrm{d}B}{\mathrm{d}t} = -2\frac{\mu\mathrm{T}}{\mu\mathrm{s}} = -2\frac{\mathrm{T}}{\mathrm{s}}$:

$$|\varepsilon| = \pi(0.16m)^2 \left|-2\frac{\mathrm{T}}{\mathrm{s}}\right| = 0.16\frac{\mathrm{m}^2\mathrm{T}}{\mathrm{s}}$$

$$|\mathrm{emf}| = 0.16V$$

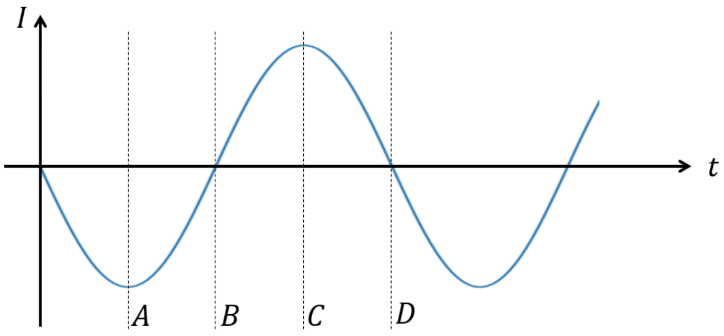
Words: 53

2. Energy transfer in LC circut

2.1. Specify Question

Question 7: Energy transfer in LC circuit

a) A charged capacitor, C, is connected in series with an inductor, L. The current in the circuit as a function of time is shown below. At which point(s) is the energy in the inductor the largest? Select all that apply.



- [ ] A
- [x] B ✕
- [ ] C
- [x] D ✕

Select all possible options that apply. ?

✕ 0%

b) If $C = 20\,\mathrm{pF}$ and $L = 150\,\mathrm{mH}$, what is the period T with which the **energy** oscillates between the capacitor and inductor?

$T =$ 3.75 s ? ✕ 0%

Correct answer

A
C

$T =$ `5.441398092702652e-06 second`

2.2. Diagnosis Phase

For part a, I mistook that voltage refers to energy itself, and not potential energy. Under this assumption, when we know that the inductor has peak energy when there is the least energy in the wires.

I forgot the equation to relating L, C, and the period of current in the LC circuit just put down something random.

$$\frac{150\ \mathrm{mH}}{20\ \mathrm{pF}}\left(\frac{1}{2}\right) = 3.75$$

Words: 56

2.3. Correction Phase

The equation I forgot was:

But the relationship I needed was, in an oscillating LC circuit:

$$\omega = \sqrt{\frac{1}{LC}}$$

Then using the relationship between ω and the period:

$$\omega = \frac{2\pi}{T}$$
$$\frac{1}{\sqrt{LC}} = \frac{2\pi}{T}$$
$$T = 2\pi\sqrt{LC}$$

Therefore:

$$T = 2\pi\sqrt{LC}$$

But this is the period of the oscillations in charge, and we want the oscillations in energy. Comparing this to a ball on spring system with the energy being transferred from spring potential to kinetic, the period of energy oscillation is half the period of position oscillation. Making this analogy:

$$T_{\text{energy}} = \frac{T_{\text{charge}}}{2} = \pi\sqrt{LC} = \pi\sqrt{(150\times10^{-3}\ \mathrm{H})(20\times10^{-12}\ \mathrm{F})} = 0.000005441398093\ \mathrm{s}$$
$$\therefore T_{\text{energy}} = 5.44\times10^{-6}\ \mathrm{s}$$

Words: 75

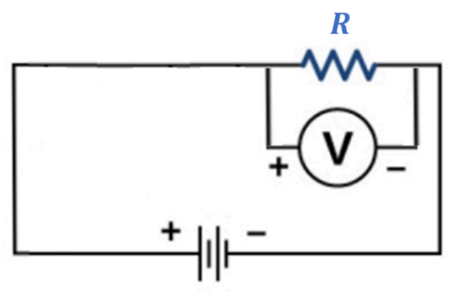
3. Measuring current with a voltmeter

3.1. Specify Question

Question 8: Measuring current with a voltmeter

A $9\,\text{V}$ battery with internal resistance of $13\,\Omega$ is connected to a load resistor with resistance $R = 160\,\Omega$ (shown in blue). The potential difference across the load resistor is $0.34\,\text{V}$ as measured by a voltmeter with a $10\,\text{M}\Omega$ internal resistance. Note that the internal resistances of the battery and voltmeter are not shown in the diagram below. What current is flowing through the load resistor?

Answer in mA to the nearest $0.1\,\text{mA}$.



$I =$ 2.125 mA ✕ 0%

Correct answer

$I = 2.1\,\text{mA}$

3.2. Diagnosis Phase

Clerical Error, I forgot to round the result to the nearest 0.1 mA

Words: 13

3.3. Correction Phase

The voltmeter has such a high internal resistance compared to the load resistance that we can treat it as negligible to simplify the calculations. So we can just treat the voltmeter as being a perfect voltmeter with infinite internal resistance.

$$V = IR$$

$$V = IR_{\text{load}}$$

$$I = \frac{V}{R_{\text{load}}} = \frac{0.34V}{160\Omega} = 0.002125A = 2.125 \text{ mA} \approx 2.1 \text{ mA}$$

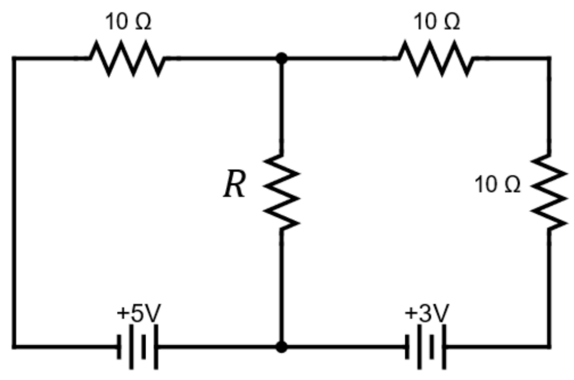
Words: 40

4. Find the current through the resistor

4.1. Specify Question

Question 1: Find the current through the resistor

What is the magnitude of the current flowing through the $R = 5\,\Omega$ resistor?



$I =$ 1 A ✕ 0%

Correct answer

$I = 0.2$ ampere

4.2. Diagnosis Phase

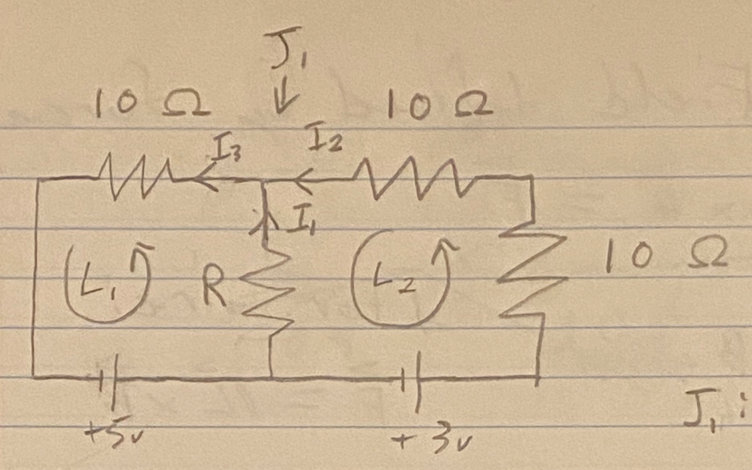
I was going too fast and made a mistake in my calculations and solving of the linear system which somehow made the current come out to 1 Amp.

I also remember forgetting the direction that the batteries were facing, thinking that both batteries were facing in the opposite direction.

Words: 49

4.3. Correction Phase

Using both Kirchhoff's Circuit Laws the junction and loop rules:



$J_1:\ I_3 = I_1 + I_2$

$L_1:\ 5v - 5I_1 - 10I_3 = 0$

$L_2:\ 3v - 10I_2 - 10I_2 + 5I_1 = 0$

$3v - 20I_2 + 5I_1 = 0$ $\qquad 5v - 5I_1 - 10(I_1 + I_2) = 0$

$5v - 15I_1 - 10I_2 = 0$

$\times -2 \downarrow$

$-10v + 20I_2 + 30I_1 = 0$

Elimination $\rightarrow$ $+$

$3v - 20I_2 + 5I_1 = 0$

$-7v + 35I_1 = 0$

$I_1 = \frac{7}{35} = \frac{1}{5} = 0.2\ A$

Words: 10